

Security+ 3.4 · Power Resiliency

Mestre CyberSec · Lesson 3.4 — Power Resiliency (4:02) · [Assistir Vídeo no YouTube](#) · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Surge protector (protege PICO de tensão — doméstico); UPS (bateria que segura minutos); generator (dias/semanas — diesel/gás). Datacenter: UPS + generator redundante.

💡 Key Exam Highlights & Critical Concepts

- UPS: minutos — dá tempo de shutdown limpo / gerador entrar
- Generator: dias/semanas — o UPS cobre a transição

🎯 Exam Tips & Typical CompTIA Scenarios

- 'temporary power during electrical outage'=UPS

Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 3.4 — Power Resiliency)

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Uninterruptible Power Supply (UPS)

Battery-backed power source that provides temporary electricity (typically minutes) during an outage, allowing for clean shutdown or until a secondary source engages.

 **Exam Trap:** Exam trigger 'temporary power during electrical outage' = UPS. Remember: UPS bridges the gap, generator sustains long-term.

• Surge Protector (N/A)

Device that diverts small overvoltage spikes (transients) away from connected equipment to prevent damage from brief voltage increases.

 **Exam Trap:** Do not confuse Surge (longer over-voltage) with Spike (instantaneous). Surge protector = picos PEQUENOS, domésticos, não substitui UPS.

• Generator (N/A)

Long-duration alternative power source (diesel, gas, propane) capable of sustaining facility operations for days or weeks during extended outages.

 **Exam Trap:** Generator requires seconds/minutes to spin up (ATS bridges this). On exam: extended outage = generator; UPS handles the transition window.

• Automatic Transfer Switch (ATS)

Device that automatically senses power loss and switches the electrical load from primary utility power to the backup generator (or UPS).

 **Exam Trap:** ATS is the bridge between UPS and Generator. On exam questions about 'seamless failover to generator' — look for ATS in the answer.

• Power Distribution Unit (PDU)

Rack-mounted device with multiple outlets that distributes conditioned and protected power to servers and networking equipment in a datacenter.

 **Exam Trap:** PDU ≠ surge strip. A managed PDU provides remote monitoring and outlet switching — distinguish from basic rack power strips on the exam.

• Dual Power Supplies (N/A)

Two independent power supply units installed in a single device, each connected to different circuits (often different PDUs) so failure of one does not cause downtime.

 **Exam Trap:** Exam scenario: 'failover power supply' or 'redundant PSU' = dual power supplies. Combined with separate PDUs/UPS units, this provides redundancy.

• Power Redundancy (N/A)

Datacenter design using duplicate, independent power paths (multiple UPS units, generators, PDUs, circuits) so that no single failure interrupts operations.

 **Exam Trap:** Exam loves: 'datacenter resilient design' = UPS + redundant generator + dual PSUs + multiple PDUs. Single points of failure are the wrong answer.

• Spike vs Surge (N/A)

Spike = instantaneous, very high over-voltage (nanoseconds). Surge = longer-duration over-voltage (milliseconds to seconds). Both can damage hardware.

⚠ Exam Trap: Trap question: a Surge Protector handles small surges/spikes but NOT sustained outages — for outages you still need a UPS. Don't pick surge protector when the question says 'outage'.

✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A small accounting firm has a file server and VoIP gateway. Utility power often flickers for 5–30 seconds during storms. There is no backup generator. The firm wants the devices to ride through brief outages and shut down gracefully during longer ones. Which solution best meets this need?

- A) A rack-mounted surge protector with network filtering.
- B) An uninterruptible power supply sized for the critical load.
- C) An automatic transfer switch connected to the municipal feed.
- D) A redundant power distribution unit fed by two wall circuits.

QUESTION 2 · EXAM SCENARIO

A data center has a diesel generator and a UPS. When utility power fails, the generator takes 45 seconds to start and stabilize. The UPS batteries currently provide only 8 minutes of runtime. Which component is responsible for automatically disconnecting from the utility feed and connecting the generator to the building's critical load once generator power is stable?

- A) Automatic transfer switch (ATS)
- B) Power distribution unit (PDU)
- C) Surge protector
- D) Dual power supply

QUESTION 3 · EXAM SCENARIO

An administrator configures a rack server with two power inlets. Both power cords are plugged into the same PDU, which is connected to one UPS. During a UPS maintenance test, the server powers off. Which change best increases power redundancy for this server?

- A) Replace the server's power supplies with higher-wattage models.
- B) Connect each power inlet to a different PDU powered by separate UPS systems.
- C) Add a surge protector between the server and the PDU.
- D) Configure the server to boot automatically after power returns.

QUESTION 4 · EXAM SCENARIO

A hospital's critical network closet has a generator that starts within 10 seconds after an outage. However, a network switch reboots immediately when utility power drops, causing a 10-second outage for clinical staff. Which device should be installed to prevent the switch from rebooting while the generator starts?

- A) A larger circuit breaker
- B) A second generator with manual start
- C) A metered power distribution unit
- D) A battery-backed uninterruptible power supply

QUESTION 5 · EXAM SCENARIO

A workstation in an older office building repeatedly suffers damage to its power supply after lightning storms. The user does not need to keep working during a blackout. Which device is most appropriate to protect the workstation from high-voltage transients?

- A) Uninterruptible power supply
- B) Standby generator
- C) Surge protector
- D) Automatic transfer switch

QUESTION 6 · EXAM SCENARIO

A power quality monitor records an event lasting a few microseconds in which voltage briefly rises to several thousand volts after a nearby lightning strike. This event destroys a network interface card. How should the technician classify this event in the incident report?

- A) Blackout
- B) Brownout
- C) Surge
- D) Spike

QUESTION 7 · EXAM SCENARIO

A company wants to ensure that a single UPS failure cannot take down its most critical database server. The server has two power supplies. The company already has two independent utility feeds and two UPS systems. Which cabling design best achieves this goal?

- A) Both power supplies connected to one PDU on UPS A.
- B) One power supply on UPS A and the other on UPS B, using separate PDUs.
- C) Both power supplies connected to UPS A, with UPS B reserved as a spare.
- D) One power supply connected to a surge protector and the other to UPS A.

QUESTION 8 · EXAM SCENARIO

A datacenter administrator needs to remotely reboot a frozen server and monitor per-outlet power consumption in a rack without physically entering the cage. The rack currently has only basic power strips. Which device should be installed?

- A) Switched/metered power distribution unit
- B) Uninterruptible power supply
- C) Automatic transfer switch
- D) Generator

QUESTION 9 · EXAM SCENARIO

A company's backup generator is rated to run for several days, but during a prolonged outage the UPS batteries are exhausted after 12 minutes and the generator has not yet started. The generator's starter battery is dead. Which action best prevents this specific failure in the future?

- A)** Add more UPS batteries to extend runtime to several days.
- B)** Include generator starter battery and fuel system in preventive maintenance checks.
- C)** Replace the generator with a larger surge protector.
- D)** Connect all servers to a single PDU instead of dual PDUs.

QUESTION 10 · EXAM SCENARIO

A data center uses utility power, a UPS, a generator, and an automatic transfer switch. Which statement best describes the UPS's role when utility power fails and the generator is starting?

- A)** It provides long-term power for days until fuel is replenished.
- B)** It replaces the need for an automatic transfer switch.
- C)** It bridges the gap by supplying battery power until the generator is stable.
- D)** It converts diesel fuel into electrical power for the critical load.

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option B — An uninterruptible power supply sized for the critical load.

A UPS provides battery-backed runtime, allowing equipment to ride through short outages and perform a graceful shutdown during longer ones. A surge protector only mitigates voltage spikes/surges and provides no runtime. An ATS needs an alternate power source such as a generator, and a redundant PDU alone does not supply battery power.

Question 2: Option A — Automatic transfer switch (ATS)

An automatic transfer switch (ATS) monitors the primary utility feed and automatically transfers the load to the backup generator when the generator output is stable. A PDU distributes power, a surge protector suppresses transients, and dual power supplies provide server-level redundancy, not facility-level source transfer.

Question 3: Option B — Connect each power inlet to a different PDU powered by separate UPS systems.

To achieve power redundancy, the two server power inlets should be connected to separate power paths, such as different PDUs fed by different UPS systems or utility feeds. Higher-wattage power supplies, surge protectors, and auto-boot do not eliminate the single point of failure created by using one PDU/UPS.

Question 4: Option D — A battery-backed uninterruptible power supply

A battery-backed UPS provides immediate ride-through power during the interval between utility loss and generator stabilization. A larger breaker, manual-start generator, or metered PDU does not provide instantaneous battery power to prevent the switch from rebooting.

Question 5: Option C — Surge protector

A surge protector is designed to clamp or divert high-voltage spikes and surges before they reach the workstation's power supply. A UPS may include surge protection, but if no battery runtime is required, a surge protector is the most direct and appropriate control. Generators and automatic transfer switches address power outages, not transient overvoltage.

Question 6: Option D — Spike

A spike is a very brief, high-voltage transient, often lasting microseconds and capable of damaging electronics. A surge is a longer-duration overvoltage. A blackout is a complete loss of power, and a brownout is a prolonged undervoltage condition.

Question 7: Option B — One power supply on UPS A and the other on UPS B, using separate PDUs.

The server should use separate power paths for each power supply. Connecting one supply to UPS A and the other to UPS B through separate PDUs prevents a single UPS or PDU failure from removing power from both supplies. Using one UPS, one PDU, or a surge protector on one side leaves a single point of failure or unbalanced protection.

Question 8: Option A — Switched/metered power distribution unit

A switched and metered PDU provides per-outlet power control and monitoring, enabling remote reboots and power consumption visibility. A UPS provides battery runtime, an ATS transfers between power sources, and a generator provides long-term backup power; none of those directly provide remote per-outlet switching and metering.

Question 9: Option B — Include generator starter battery and fuel system in preventive maintenance checks.

The failure was caused by a dead generator starter battery, which prevented the generator from starting. Routine preventive maintenance of the generator, including its starter battery and fuel system, is the appropriate control. Adding UPS batteries could mask the issue but is not the specific fix, and the other options do not address generator starting reliability.

Question 10: Option C — It bridges the gap by supplying battery power until the generator is stable.

The UPS supplies battery power during the brief interval between utility loss and generator stabilization. The generator provides long-term power, the ATS transfers the load, and the generator converts fuel into electricity. The UPS does not replace the ATS or run for days.